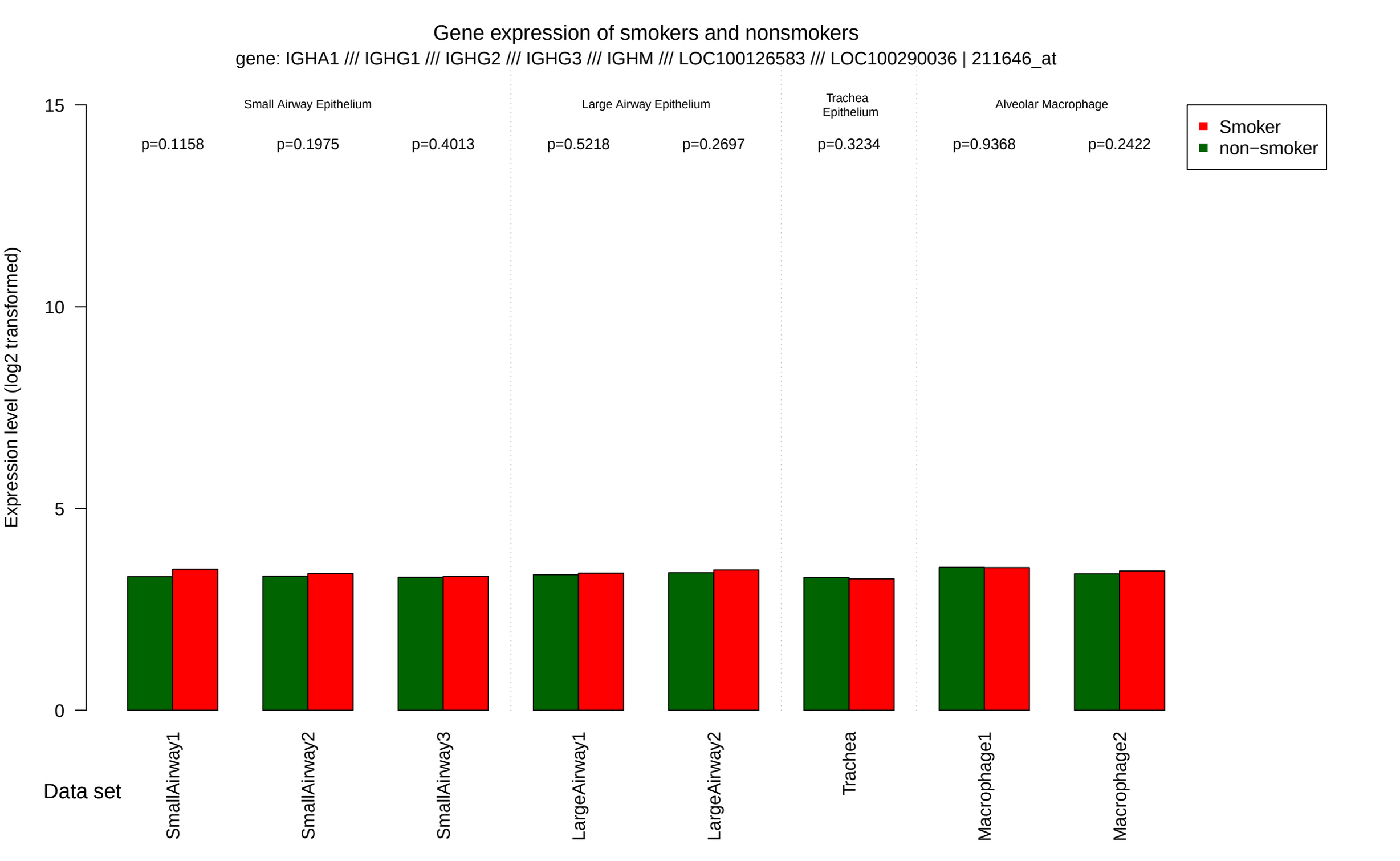
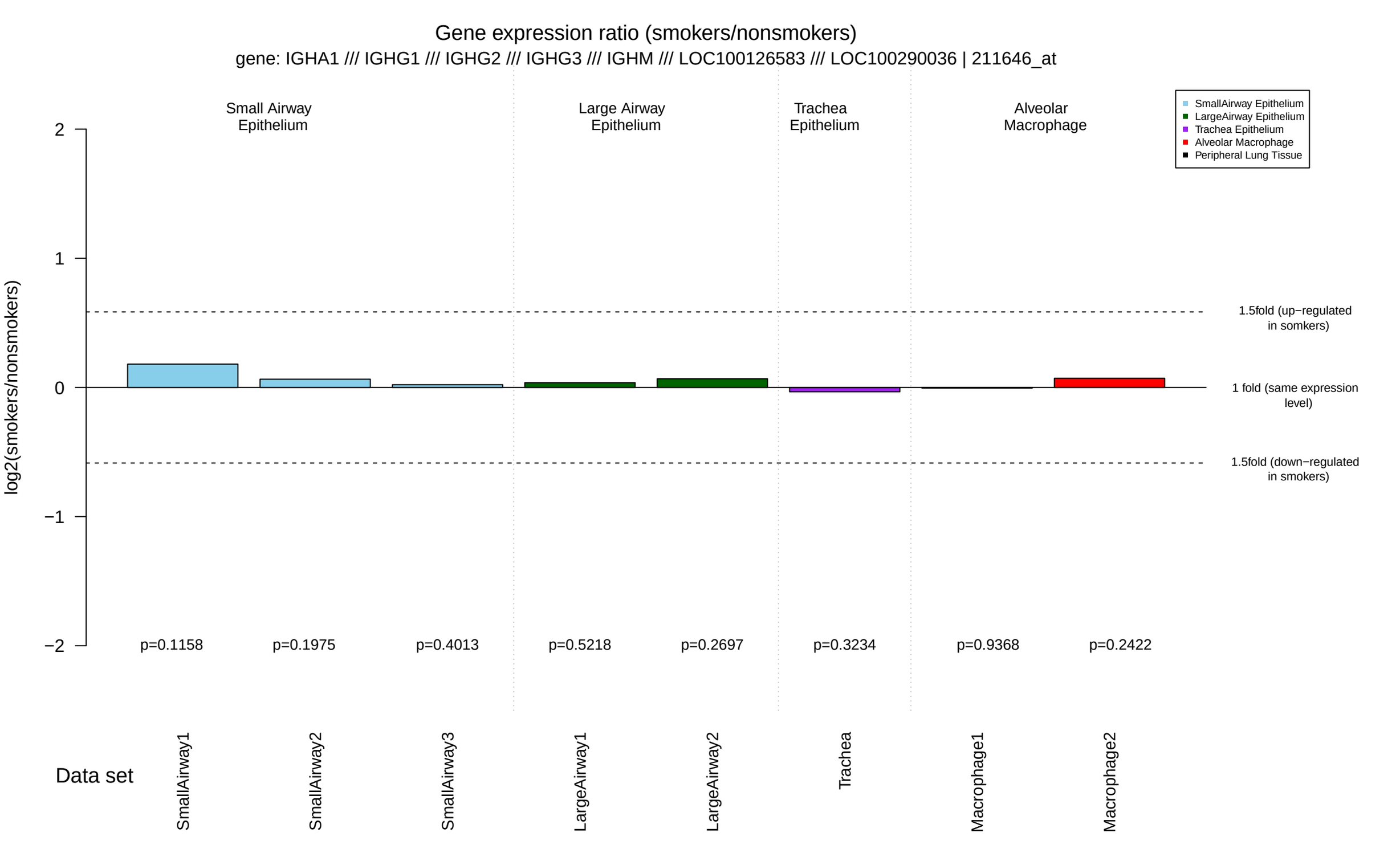
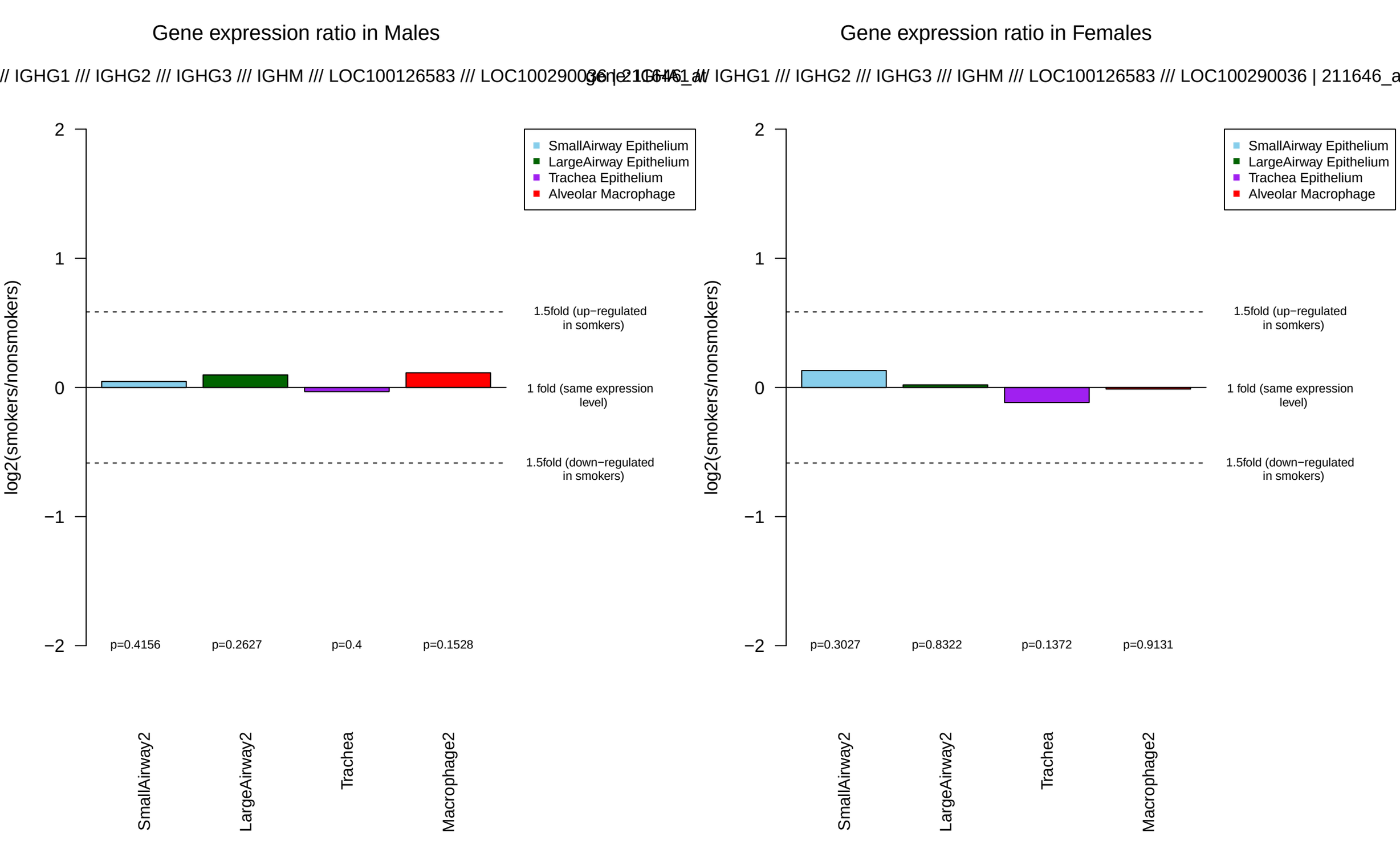
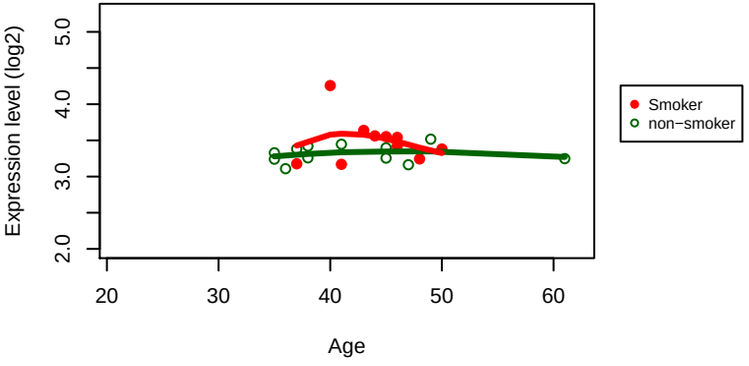

Gene expression of smokers and nonsmokers
gene: IGHA1 /// IGHG1 /// IGHG2 /// IGHG3 /// IGHM /// LOC100126583 /// LOC100290036 | 211646_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1158
p=0.1975
p=0.4013
p=0.5218
p=0.2697
p=0.3234
p=0.9368
p=0.2422
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2


Gene expression ratio (smokers/nonsmokers)
gene: IGHA1 /// IGHG1 /// IGHG2 /// IGHG3 /// IGHM /// LOC100126583 /// LOC100290036 | 211646_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1158
p=0.1975
p=0.4013
p=0.5218
p=0.2697
p=0.3234
p=0.9368
p=0.2422
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2


Gene expression ratio in Males
// IGHG1 /// IGHG2 /// IGHG3 /// IGHM /// LOC100126583 /// LOC100290036 | 211646_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4156
p=0.2627
p=0.4
p=0.1528
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
// IGHG1 /// IGHG2 /// IGHG3 /// IGHM /// LOC100126583 /// LOC100290036 | 211646_a
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3027
p=0.8322
p=0.1372
p=0.9131
SmallAirway2
LargeAirway2
Trachea
Macrophage2

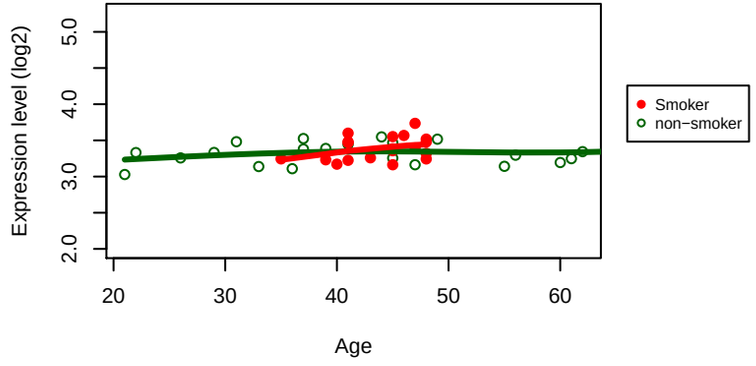
set: SmallAirway1 (p=0.1158)

/// IGHG2 /// IGHG3 /// IGHM /// LOC100126583 /// LOC100290036 | 21



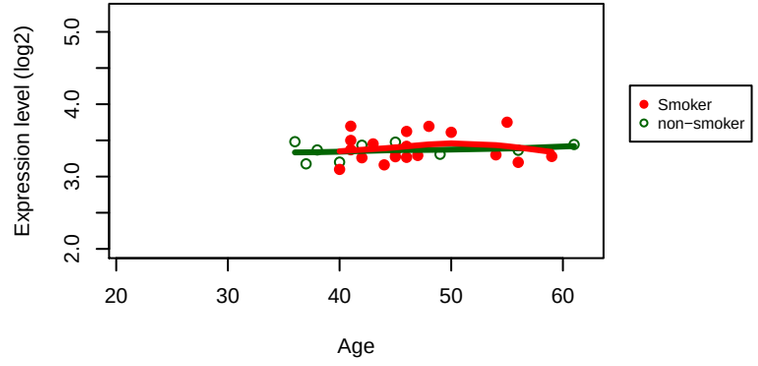
set: SmallAirway2 (p=0.1975)

/// IGHG2 /// IGHG3 /// IGHM /// LOC100126583 /// LOC100290036 | 21



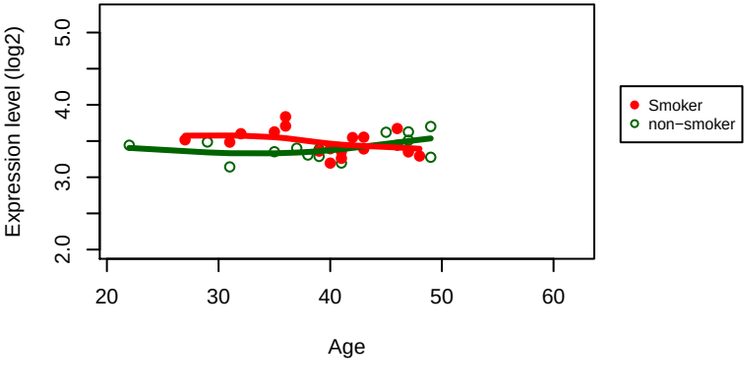
set: LargeAirway1 (p=0.5218)

/// IGHG2 /// IGHG3 /// IGHM /// LOC100126583 /// LOC100290036 | 21



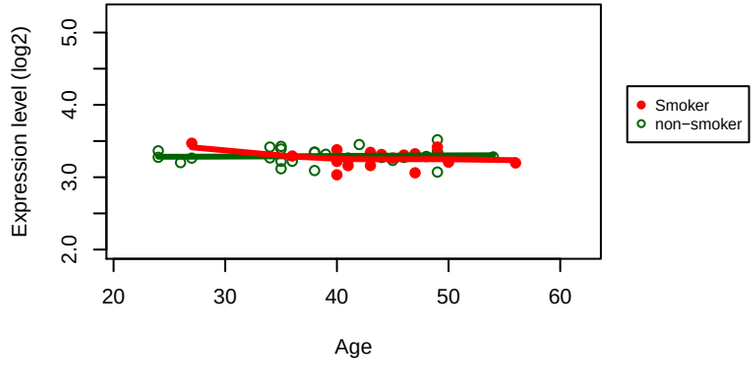
set: LargeAirway2 (p=0.2697)

/// IGHG2 /// IGHG3 /// IGHM /// LOC100126583 /// LOC100290036 | 21



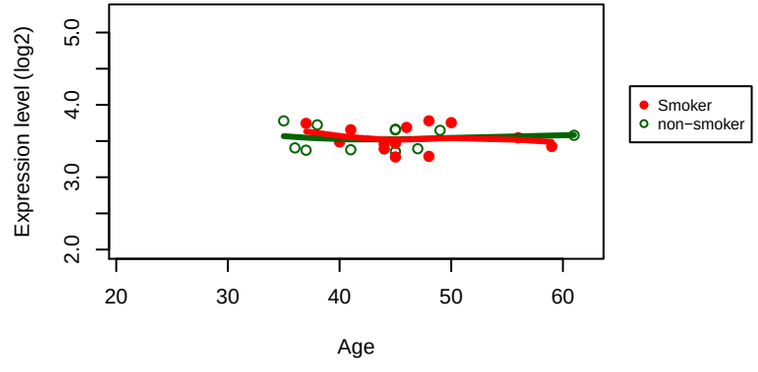
set: Trachea (p=0.3234)

/// IGHG2 /// IGHG3 /// IGHM /// LOC100126583 /// LOC100290036 | 21



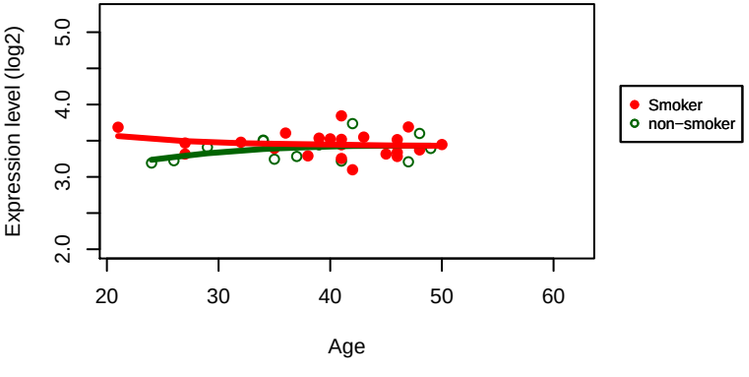
set: Macrophage1 (p=0.9368)

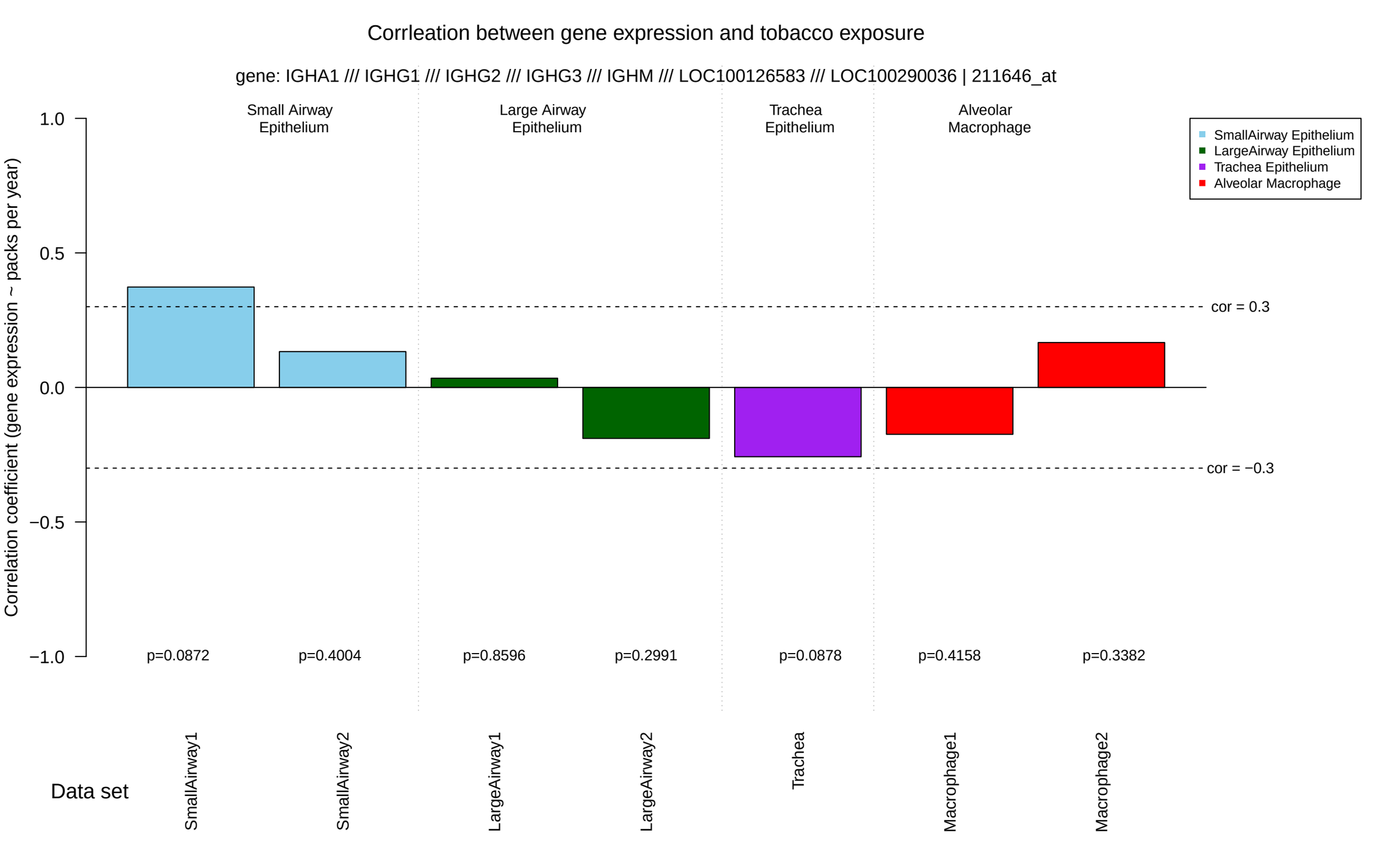
/// IGHG2 /// IGHG3 /// IGHM /// LOC100126583 /// LOC100290036 | 21



set: Macrophage2 (p=0.2422)

/// IGHG2 /// IGHG3 /// IGHM /// LOC100126583 /// LOC100290036 | 21




Corrleation between gene expression and tobacco exposure
gene: IGHA1 /// IGHG1 /// IGHG2 /// IGHG3 /// IGHM /// LOC100126583 /// LOC100290036 | 211646_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0872
p=0.4004
p=0.8596
p=0.2991
p=0.0878
p=0.4158
p=0.3382
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2